

Early Warning System for ICU Sepsis Prediction

What is Sepsis?

Sepsis is a dangerous condition where the body's immune system overreacts to an infection and starts damaging its own organs. It can lead to kidney failure, liver damage, heart problems, and even death. Early detection can save lives.

Problem Statement

ICU patients generate large amounts of medical data every hour. Doctors may miss small changes that indicate sepsis. The goal is to use machine learning to predict sepsis before it becomes obvious.

Goal

Build a model that predicts whether a patient is likely to develop sepsis in the next few hours.

Input

- Heart Rate
- Blood Pressure
- Temperature
- Respiratory Rate
- Oxygen Saturation (SpO₂)
- Laboratory Test Results
- Age
- Gender

Output

- No Sepsis
- Sepsis Likely
- Or a probability score (e.g., Sepsis Risk = 91%)

Dataset

PhysioNet/CinC Challenge 2019 Dataset containing data from over 40,000 ICU patients.

Dataset Features

- Patient ID
- Age
- Gender
- Heart Rate
- Blood Pressure
- Temperature
- Oxygen Level
- Lab Results
- Sepsis Label (Yes/No)

Dataset Availability

Publicly available from the PhysioNet portal.

Learning Type

Supervised Learning because each patient record includes the correct label (Sepsis or No Sepsis).

Data Mining Process

- Data Cleaning: Remove missing values, incorrect values, and duplicates.
- Time-Series Feature Extraction: Use trends over time instead of single measurements.
- Feature Selection: Keep only the most important medical features.
- Handle Class Imbalance: Use SMOTE to balance the dataset.
- Train Models: Use algorithms such as Random Forest and Gradient Boosting.
- Explain Predictions: Use SHAP to explain why a prediction was made.

Workflow

Dataset → Cleaning → Feature Engineering → Feature Selection → Train Model → Predict Sepsis → Explain Result (SHAP)

Advantages

- Large dataset
- Real healthcare problem
- High impact
- Beginner friendly

Challenges

- Missing values
- Imbalanced classes
- Time-series data

Unsupervised Anomaly Detection for Cardiovascular Arrhythmia

What is Arrhythmia?

Arrhythmia means an abnormal heartbeat. The heart may beat too fast, too slow, or irregularly. Serious cases can lead to stroke, heart failure, or sudden cardiac death.

Problem Statement

Continuous ECG monitoring produces thousands of heartbeats for each patient, making manual inspection difficult. The goal is to automatically detect unusual heartbeats.

Goal

Build a system that automatically identifies abnormal heartbeats.

Input

- ECG waveform
- Heart Rate
- RR Interval
- QRS Width

Output

Normal heartbeat, Anomaly, or an anomaly score (e.g., 0.91).

Dataset

MIT-BIH Arrhythmia Database containing 48 ECG recordings with thousands of annotated heartbeats.

Dataset Features

- ECG waveform
- Beat position
- Heart rate
- RR Interval
- QRS Duration

Dataset Availability

Publicly available through PhysioNet.

Learning Type

Unsupervised Learning. The model learns what normal heartbeats look like and flags unusual patterns as anomalies.

Data Mining Process

- Signal Preprocessing: Remove noise, motion artifacts, and baseline drift.
- Feature Extraction: Convert ECG signals into numerical features.
- Dimensionality Reduction: Use PCA and t-SNE (mainly for visualization).
- Anomaly Detection: Use Isolation Forest, One-Class SVM, or DBSCAN.

Workflow

ECG Signal → Noise Removal → Feature Extraction → Dimensionality Reduction → Anomaly Detection → Normal / Abnormal Beat

Advantages

- Unique project
- Uses real ECG data
- Good introduction to anomaly detection

Challenges

- Signal processing is more difficult
- ECG data can be noisy
- Requires understanding anomaly detection

Comparison

Feature	Proposal 1	Proposal 4
Problem	Predict Sepsis	Detect Arrhythmia
Input	Vital signs + Lab tests	ECG Signals
Output	Sepsis Risk	Normal/Anomaly
Dataset	PhysioNet CinC 2019	MIT-BIH Arrhythmia
Learning	Supervised	Unsupervised
Difficulty	Moderate	Moderate to High